

Technology Stack

Date	07 July 2026
Team ID	XXXXXX
Project Name	Smart Lender
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	HTML5, CSS3, JavaScript	To create a responsive and user-friendly interface for entering loan applicant details.
2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python, Flask	Flask handles user requests, processes input, loads the trained model, and returns loan predictions efficiently.
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	CSV Dataset, Pickle (.pkl) Model File	The CSV dataset is used for training, and the trained XGBoost model is stored as a Pickle file for prediction.

4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	IBM Cloud	To deploy the Smart Lender application online for easy accessibility and scalability
5	Version Control & CI/CD	<i>Git, GitHub</i>	To manage source code, track project changes, and enable team collaboration.
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn	Used for data preprocessing, visualization, model training, evaluation, and analysis.